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Supplementary information

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LRRK2-targeting antisense oligonucleotide in Parkinson's disease: a phase 1 randomized controlled trial

In the format provided by the authors and unedited

Supplementary information

Supplementary data: Preclinical study in cynomolgus macaques

Results

Safety

In total, 11 doses of intrathecal (IT) BIIB094 12 mg, 24 mg or 40 mg were administered to cynomolgus macaques (N = 32) once every 4 weeks over 41 weeks. BIIB094 was well tolerated and no adverse effects were observed in macaques receiving BIIB094 40 mg. Transient changes in clinical signs were observed with the 12 mg and 40 mg doses (mainly in two animals), as well as partially reversible cerebrospinal fluid (CSF) cell count and clinical chemistry parameter changes at doses of 12 mg or above. Non-adverse changes in histopathology and clinical pathology were observed in multiple organs at all dose levels. All were reversible except for macrophage infiltration in the brain and optic nerve, microvesicular neuronal vacuolation in the spinal cord, and minimal increases in kidney weight. Based on the general reversibility and non-severity of these changes, and the lack of any correlation with clinical signs or clinical pathology findings, BIIB094 40 mg was established as the no observable adverse event level (NOAEL).

Pharmacokinetics

BIIB094 reached peak concentration ($C_{max)}$ in serum 1–4 hours after administration in macaques (Supplementary Fig. 5A and 5B). Serum levels then declined biphasically and fell to close to, or below, the lower limit of quantification (LLOQ; 1.00 ng/mL) by 48 hours after administration. Geometric mean C_{max} and area under the curve (AUC) at the NOAEL of BIIB094 40 mg were 22,000 ng/mL and 134,000 hr*ng/mL, respectively. In animals receiving BIIB094 40 mg, mean BIIB094 levels reached approximately 60 µg/g in the frontal cortex, 1400 µg/g in the kidney cortex and 3 µg/g in the lung (Supplementary Fig. 5C–E).

Pharmacodynamics

Levels of *LRRK2* mRNA were significantly lowered at all doses studied in the frontal cortex of macaques compared with artificial CSF (aCSF) control (> 90%; p < 0.0001), but not in the kidney cortex or lung (Supplementary Fig. 5F–H).

Methods: Preclinical study in cynomolgus macaques

Animals and ethics

The study in cynomolgus macaques (*Macaca fascicularis*) was performed by Charles River Laboratories (Montreal ULC). Animals (N = 32) were 2–3 years old at experimentation start, of mixed sex (4 males, 4 females per cohort) and of Asian origin. The study was performed in compliance with Good Laboratory Practice and as accepted by Regulatory Authorities throughout the European Union, United States of America (Food and Drug Administration), Japan (Ministry of Health, Labour and Welfare), and other countries that are signatories to the Organisation for Economic Co-operation and Development (OECD) Mutual Acceptance of Data Agreement.

Dosing

Animals were food deprived at least 6 hours prior to dosing. Animals were anesthetized. Percutaneous lumbar IT dosing of BIIB094 12, 24 or 40 mg formulated in 1 mL aCSF was performed on days 1, 29, 57, 85, 113, 141, 169, 197, 225, 253 and 281. Control animals received aCSF. Injections were performed using atraumatic spinal needles (e.g., B Braun, 25 Gauge, 1 inch) and were followed by a 0.25 mL aCSF flush. Animals were anesthetized with ketamine/medetomidine anesthesia, dosed in a lateral recumbency position and remained in a prone position for 15 minutes after dosing was complete, before administration of atipamezole reversal agent.

Tissue dissection for RNA extraction and real-time quantitative reverse transcription polymerase chain reaction (qRT-PCR)

Animals were sacrificed 7 days after the last dose. Brain slices, kidney cortex and left lung lobes were collected, flash frozen on dry ice and stored frozen (–60°C to –90°C). At the time of sampling, 2 mm biopsy punches were used to collect frontal cortex from frozen brain slices and kidney cortex and lung samples for RNA extraction. Each sample was homogenized in 1 mL of Trizol reagent (Thermofisher Scientific, Waltham, MA) and total RNA was extracted according to the manufacturer's protocol. Total RNA was purified further using a mini-RNA purification kit (Qiagen, Valencia, CA). After purification, the RNA samples were subjected to real-time qRT-PCR analysis using the Life Technologies ABI QuantStudio 7 Flex Sequence Detection System (Applied Biosystems Inc, Carlsbad CA). Briefly, 10 µL RT-PCRs containing 400 nL of RNA were run with the AgPath- ID One-Step qRT-PCR Kit (Thermofisher Scientific, Waltham, MA) reagents with primer probe sets targeting monkey *LRRK2* and *PPIA*, a ubiquitously expressed housekeeping gene. Primers and probes were ordered from IDT Technologies. Primers were as follows: *LRRK2* forward primer, AGATGATTCACTCAGGTCATCAA; *LRRK2* reverse primer, CTCTCAGAAGCCAGAGAAGAAA; *LRRK2* probe, TCCCATATGAGGCATTCAGACAGCA with 5'FAM and 3'MGB chemistry; *PPIA* forward primer, TGCTGGACCCAACACAAATG; *PPIA* reverse primer, TGCCATCCAACCACTCAGTC; *PPIA* probe' TTCCCAGTTTTTCATCTGCACTGCCA with 5'FAM and 3' TAMRA chemistry. All qPCRs were run in technical triplicates; outliers resulting in a greater than 20% coefficient of variation from sample average were excluded. The expression level of total *LRRK2* mRNA was normalized to that of *PPIA* mRNA, and this was further normalized to that measured in control animals that received aCSF. *LRRK2* mRNA levels were reported as a percentage of aCSF control. Data were analyzed using Microsoft Excel. One-way ANOVA with Dunnett's multiple comparison against aCSF-treated group were used to analyze mRNA data from macaques treated with aCSF or BIIB094 (GraphPad Prism 10.4.2).

Pharmacokinetics

Serum concentrations of BIIB094 were measured using a validated hybridization enzyme-linked immunosorbent assay. The following pharmacokinetic parameters were computed: AUC_{0-48} (48 hours after dose), AUC t_{last}, AUC t_{last}/Dose, R_{AUC}, C_{max}, T_{max} and $t_{½}$.

Protocol approval

The Study Director, Charles River Laboratories (Senneville, QC) Test Facility Management (TFM) and Institutional Animal Care and Use Committee (IACUC) approved the protocol.

Supplementary methods: REASON study

Antisense oligonucleotide synthesis

The synthesis of oligonucleotides was carried out on an AKTA Process solid-phase synthesizer using the NittoPhase HL Unylinker (Kinovate Life Sciences) as the solid support at ambient temperature. The phosphoramidite starting materials (MOE-A, MOE-G, MOE-5^{Me}C, MOE-5^{Me}U, Deoxy-A, Deoxy-G, deoxy-T and deoxy-5^{Me}C) were purchased from Hongene. Solvents and reagents were procured from Tedia Company, LLC.

The 4-reaction synthesis for attaching each nucleotide consisted of detritylation, coupling, capping and sulfurization or oxidation reactions. The detritylation step was carried out using a 10% solution of dichloroacetic acid in toluene as the reagent. The coupling step was carried out by delivering equal volumes of a solution of the corresponding phosphoramidite (0.2 M in acetonitrile) and a solution of 4,5-dicyanoimidazole (1.0 M in acetonitrile containing 0.1 M N-methylimidazole). The sulfurization step was completed by delivery of 0.2 M of PADS (Phenylacetyl disulfide) in acetonitrile/3-picoline (1:1, v/v) (the PADS solution was aged for at least 12 h before use according to the literature procedure)[1]. The oxidation was performed by delivering I_2 solution (0.05 M in pyridine-H2O, 9:1, v/v) to the column. The capping step is performed after the sulfurization or oxidation steps. The capping step was performed by delivery of the same amount of a solution of N-methyl imidazole−pyridine−ACN (2:3:5, v/v/v) and a solution of acetic anhydride in ACN (1:4, v/v) to the solid support. After each reaction, the solid support was washed with acetonitrile. After the synthesis was completed, the phosphorus protecting group was removed by circulating a mixture of Triethylamine and acetonitrile (1:1, v/v). Cleavage of the oligonucleotides from the support and base deprotection were performed in ~30% NH_4OH at 50−60°C in a stainless-steel vessel to afford the crude DMT-on oligonucleotide product.

The crude DMT-on intermediate is purified by reversed-phase high performance liquid chromatography (RP-HPLC) using a polymer stationary phase and aqueous sodium acetate/methanol mobile phases. The DMT-on product peak is collected, and the other peaks are discarded.

The Trityl eluate obtained is precipitated with ethanol, isolated by sedimentation and decantation, and reconstituted in purified water to reduce levels of methanol and sodium acetate. Reconstituted 5′-O-DMT-on material is detritylated by addition of glacial acetic acid for 5-10 h and the reaction is quenched with 5 N sodium hydroxide (NaOH). The resulting detritylated material is precipitated and reconstituted two more times. Finally, the solution of purified oligonucleotide is transferred into freeze drying trays and placed in a freeze dryer. The material is frozen and then dried under reduced pressure to afford the amorphous Drug Substance.

Protein preparation and digestion for liquid chromatography mass spectrometry (LC-MS) analysis for biomarker detection

Equal 500 µL volumes of each CSF sample were denatured with 1X RIPA lysis buffer (Millipore Sigma 20-188). Samples were then spiked with 2 pg/mL of LRRK2 stable isotope labeled peptide $K_{13C(6)15N(2)}$ $AEEGDLLVNPDQPR_{13C(6)15N(2)}$ and 2 pg/mL of pRab10 T72 stable isotope labeled peptide $FHTITTSYYR_{13C(6)15N(2)}$ as internal standards. Samples were then digested for 1.5 hours at 37˚C with 10 µg of TPCK treated trypsin (Millipore Sigma T1426). Standard curve samples were prepared by adding LRRK2 recombinant protein (Thermo Fisher A15197) and pRab10 T72 recombinant protein (kindly provided by Dario Alessi, University of Dundee) to 500 µL of 0.1% bovine serum albumin surrogate matrix followed by addition of 1X RIPA lysis buffer and internal standards. Standard curve intervals were constructed at 1, 2, 5, 10, 25, 50, 75, 100 pg/mL for LRRK2 and 0.2, 0.4, 1, 2, 5, 10, 15, 20 pg/mL for pRab10.

Immunoprecipitation of LRRK2 and pRab10 target peptides

Immunoprecipitation of LRRK2 and pRab10 target peptides was performed on an Agilent AssayMAP Bravo. First, N241A/34 LRRK2 antibody (Antibodies Inc. 75-253) and MJF-R20 pRab8A phospho T72 antibody (Abcam ab230260) were labeled with ChromaLINK biotin (Vector Labs B-1001). Antibodies were then immobilized on AssayMAP 5 µL streptavidin SA-W cartridges (Agilent Technologies G5496-60010) at a ratio of 1 µg of each antibody per cartridge. Digested sample was then loaded onto each cartridge at a flow rate of 10 µL/min followed by a PBS wash. Samples were then eluted with 50 µL of 2% acetonitrile in water acidified with 0.1% formic acid.

LC-MS analysis of LRRK2 and pRab10 target peptides

Eluted samples were loaded onto an Evotip Pure and chromatographically separated using an Evosep One running the 40SPD method and coupled to a Thermo EASY-Spray PepMap RSLC C18 column (Thermo Scientific ES904). Mass spectrometry was performed on a Thermo Exploris 480 operating in parallel reaction monitoring (PRM) mode. Targets included m/z 560.9566 $[M+3H]^{3+}$ (native LRRK2 peptide), m/z 566.9641 $[M+3H]^{3+}$ (LRRK2 internal standard), m/z 456.8710 $[M+3H]^{3+}$ (native pRab10 T72 phosphopeptide) and m/z 460.2071 $[M+3H]^{3+}$ (pRab10 T72 internal standard). PRM scans were collected at a resolution of 120,000 with automatic gain control (AGC) target set to standard ($1x10^6$) and maximum injection time set to automatic detection. Higher collisional dissociation (HCD) collision energy was optimized at 20% and 25% for the LRRK2 and pRab10 peptide pairs, respectively. Quantitation and data analysis were performed in SkyLine 24.1.0.199.

Recombinant and CSF protein preparation and digestion for LC-MS analysis

A multiplexed LC-MS method for the detection of 12 lysosome-related proteins in human CSF was developed. Equal 25 µL volumes of each CSF sample were spiked into equal 25 µL volumes of artificial CSF (aCSF; 125 mM NaCl, 26 mM $NaHCO_3$, 1.25 mM NaH_2PO_4, 2.5 mM KCl). Samples were then spiked with 2 pg/mL of isotope labeled peptides TITLEVEPSDTIENV**K**$_{13C(6)15N(2)}$ for ubiquitin, LLMLDDQ**R**$_{13C(6)15N(2)}$ for glucocerebrosidase, AYVPIAQV**K**$_{13C(6)15N(2)}$ for GPNMB, IPLNDLF**R**$_{13C(6)15N(2)}$ for LAMP2, AFSVNIF**K**$_{13C(6)15N(2)}$ for LAMP1, EVAGLWI**K**$_{13C(6)15N(2)}$ for GM2A, IPVIIE**R**$_{13C(6)15N(2)}$ for LC3, VSTLPAITL**K**$_{13C(6)15N(2)}$ for cathepsin D, FSDLTEEEF**R**$_{13C(6)15N(2)}$ for cathepsin F, DIMAEIY**K**$_{13C(6)15N(2)}$ for cathepsin B, NSWGEEWGMGGYV**K**$_{13C(6)15N(2)}$ for cathepsin L and GIDSDASYPY**K**$_{13C(6)15N(2)}$ for cathepsin S, as internal standards. Samples were then spiked with 25 µL 50:50 acetonitrile (OmniSolv® LC-MS grade, AX0156) and water (OmniSolv® LC-MS grade, WX0001), 8 µL of 0.5 mg/mL L-glutathione (Sigma Aldrich, G4251) and 25 µL of 150 µg/mL bulk trypsin (Worthington bulk trypsin TPCK Treated CAT#LS003741) in a neat artificial CSF. Samples were then digested overnight at 37˚C. Standard curve samples were prepared by adding recombinant protein for ubiquitin (Boston Biochem U-100H), glucocerebrosidase (Abcam ab235716), GPNMB (Novus NBP2-23082), LAMP1 (R&D systems 4800-LM), LAMP2 (R&D systems 6228-LM), GM2A (Abcam ab178464), LC3 (Novus NBP1-50960), cathepsin B (Biolegend 557704), cathepsin D (Biolegend 556702), cathepsin F (Sigma SRP0290), cathepsin L (Milipore 219402-2503) and cathepsin S (Abcam AB285884) to aCSF + 0.05% bovine serum albumin surrogate matrix followed by addition of internal standard, acetonitrile:water (50:50), 0.5 mg/mL glutathione reduced and trypsin. Standard curves were constructed using the following concentration intervals: ubiquitin, GM2A and cathepsin D: 25, 50, 100, 150, 250, 500, 750 and 1000 ng/mL; GPNMB and LC3: 0.25, 0.375, 0.5, 1.25, 2.5, 5, 12.5 and 25 ng/mL; glucocerebrosidase: 1.25, 1.88, 2.5, 6.25, 12.5, 25, 62.5 and 125 ng/mL; LAMP1: 5, 10,20,0,50,100,150 and 200 ng/mL; LAMP2: 62.5, 125, 250, 375, 625, 1250, 1875 and 2500 ng/mL; cathepsin B: 2.5, 3.75, 5, 12.5, 25, 50, 125 and 250 ng/mL; cathepsin F: 5, 7.5, 10, 25, 50, 100, 250 and 500 ng/mL; cathepsin L: 10, 20, 40, 60, 100, 200, 300 and 400 ng/mL; cathepsin S: 1, 2.5, 3.75, 5, 12.5, 25, 50 and 125 ng/mL.

Solid phase extraction

Solid phase extraction was performed using Oasis HLB 96-well µElution Plate (Waters 186001828BA). The plate was conditioned with 500 µL of 0.1% formic acid in acetonitrile, followed by equilibration using 500 µL of 0.1% formic acid in water. The 130 µL sample was loaded onto the Oasis HLB 96-well µElution Plate, followed by a wash with 500 µL of 5% methanol in water acidified with 0.1% formic acid. The peptides were eluted using 25 µL of 35% acetonitrile in water acidified with 0.1% formic acid. The eluent sample was diluted to 250 µL with 0.1% formic acid in water.

LC-MS analysis of lysosomal panel target peptides

Diluted samples (20 µL) were loaded onto an Evotip Pure and chromatographically separated using an Evosep One coupled to V1064 analytical column (100 µm internal diameter, 3 µm beads, 8 cm long, EV1064). Mass spectrometry was performed on a Thermo Exploris 240 operating in PRM mode. The system was set to positive mode with a resolution of 12,500, AGC target at 5×10^5, maximum injection time at 55 ms, and isolation window at 2 m/z. Supplementary Table 5 specifies the m/z values and optimized HCD collision energy settings for both native and internal standards. Quantitation and data analysis were performed in SkyLine 24.1.0.199.

REASON study protocol approval

Institutional review boards and ethics committees that approved the REASON study protocol included: Agencia Española del Medicamento y Productos Sanitarios (Spain); CEIC Hospital General Universitario Gregorio Marañon; Cleveland Clinic IRB; Copernicus Group IRB; McGill University Health Center-Research Ethics Board; Medicines and Healthcare products Regulatory Agency (MHRA); Ministry of Health (Israel); Northwestern University Institutional Review Board; NRES Committee South Central - Berkshire B; REK Sør-øst; Statens legemiddelverk; Tel Aviv Sourasky; WIRB-Copernicus Group (WCG) IRB; Western Institutional Review Board (WIRB).

Participant compensation

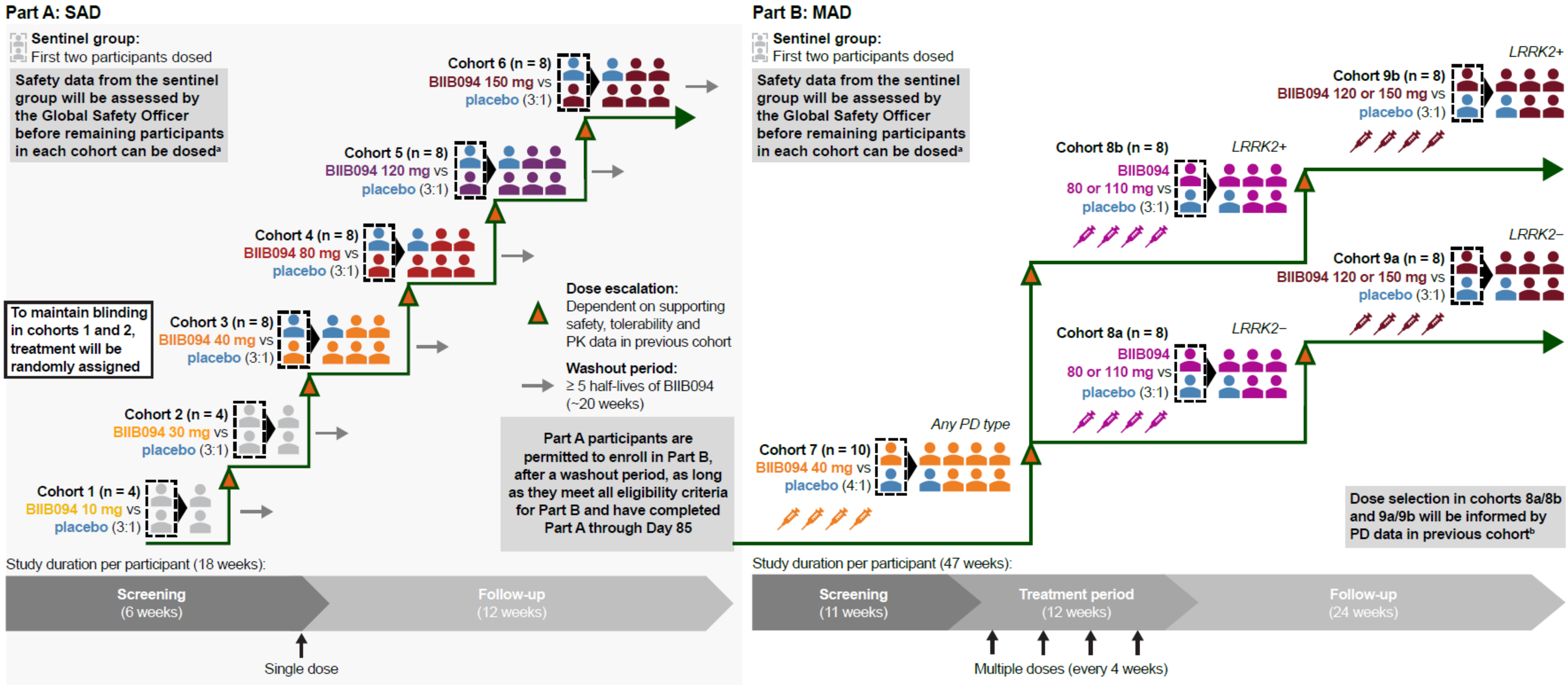
Participants in this study received compensation for their time and involvement. All study-related drugs, examinations, and medical care were provided at no cost to participants. In addition, participants were reimbursed for reasonable travel, parking, lodging, and meal expenses incurred to attend study visits. A stipend of modest value per protocol-required visit was provided for specified study visits, and an additional modest value stipend was provided for lumbar puncture procedures performed after the study treatment administration period. Reimbursements and stipends were administered in accordance with local regulations and ethics committee approval, with the option for participants to receive payments via a secure debit card system. Travel assistance was also available for participants requiring support in arranging transportation to study sites.

Supplementary References

1. Krotz, A. H., *et al.* Phosphorothioate oligonucleotides with low phosphate diester content: Greater than 99.9% sulfurization efficiency with ″aged″ solutions of phenylacetyl disulfide (PADS). *Org Process Res Dev* **8**, 852-858 (2004).

Supplementary figures

Supplementary Figure 1. Study design

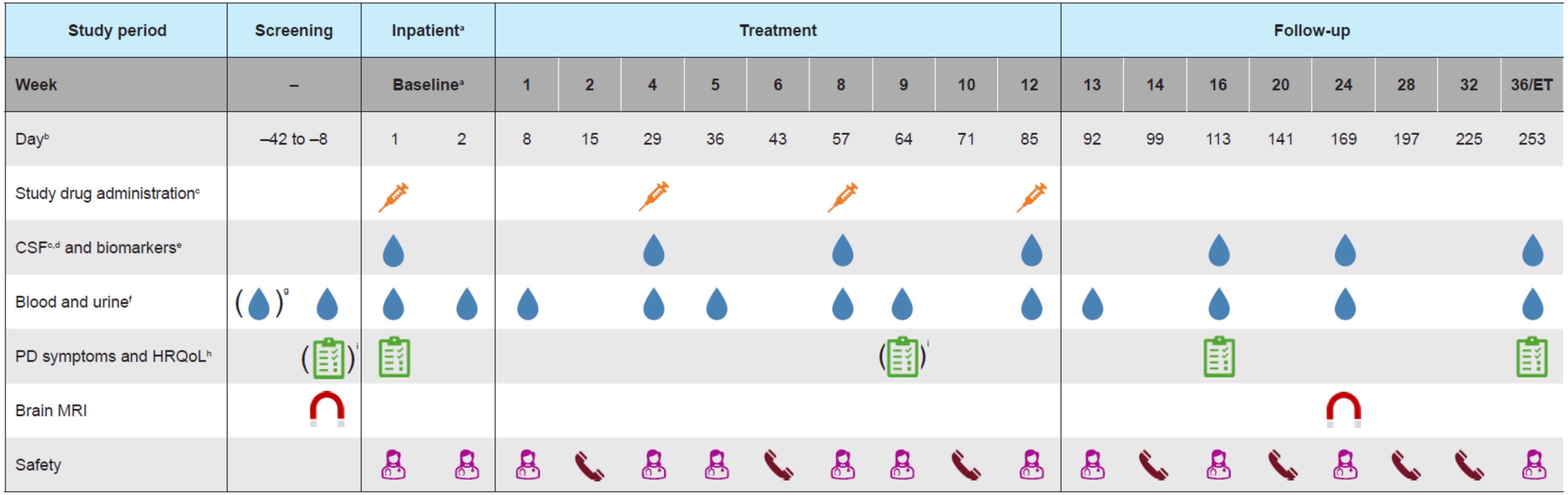


The first two participants in each cohort were dosed as a sentinel group, with one participant receiving BIIB094 and one participant receiving placebo. Safety data for the sentinel participants were reviewed by the sponsor and investigators before dosing of subsequent participants in the cohort. Progression of the trial to the next dose-level cohort was based on a blinded review of safety and pharmacokinetic data from the preceding cohorts by the sponsor and investigators.

[a]The 72-hour safety data from the first dose were assessed by the Global Safety Officer. Dosing for all non-sentinel participants was staggered approximately 24 hours apart to allow for review of safety data. [b]The safety surveillance team approval of safety and PK data from cohort 8a opened cohort 9a, and approval of safety and PK data from cohort 8b opened cohort 9b. There was a final safety/PK review for cohorts 9a and 9b.

All-PD, all PD patients (irrespective of variant status); *LRRK2*−, *leucine-rich repeat kinase 2* gene negative; *LRRK2*+, *leucine-rich repeat kinase 2* gene positive; *LRRK2*-PD, *LRRK2* variant associated PD; Non *LRRK2*-PD, non *LRRK2* variant associated PD; PD, Parkinson's disease; PK, pharmacokinetic.

Supplementary Figure 2. Schedule of assessments in Part B



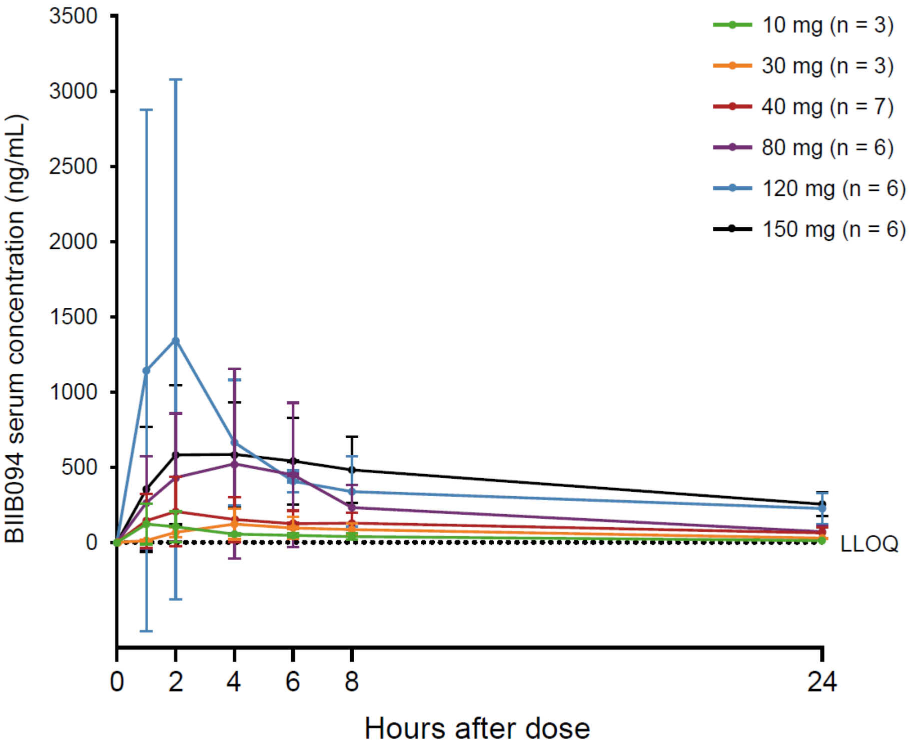
[a]Admission to clinic and baseline assessments on day of dosing or day before dosing (except pregnancy test, vital signs, electrocardiogram and lumbar puncture on day of dosing only); discharge from clinic ≥ 24 hours after dosing. [b]Visit window ±2 days in weeks 1–14 and ±3 days thereafter. [c]Lumbar puncture procedure followed by safety monitoring (telephone calls at 48 and 72 hours after the 24-hour inpatient visit; telephone calls at 24, 48 and 72 hours after outpatient visits. [d]CSF sampling performed pre-dose for pharmacokinetics, pharmacodynamics, biomarkers and safety. [e]Plasma, serum, urine and peripheral blood mononuclear cell samples. [f]Serum for pharmacokinetics and blood for safety laboratory assessments (hematology, blood chemistry, coagulation), in addition to biomarker samples (plus pregnancy testing using serum at screening and urine at weeks 4, 8 and 12 and extra tests at screening including genetic testing). [g]Screening visit 1 ≤ 35 days before main screening visit only for participants enrolling into cohorts with defined *LRRK2* variant status and no previous documentation of this status. [h]MDS-UPDRS, modified Hoehn and Yahr Scale, Modified Schwab and England ADL scale, QMA, MoCA, SCOPA-Aut, PDQ-39, ABC-16 and EQ-5D. [i]Modified Hoehn and Yahr Scale and MoCA only. [j]MDS-UPDRS, MoCA and ABC-16 only. [k]Vital signs, electrocardiography, physical and neurological examinations, laboratory safety assessments and C-SSRS (plus height at screening, weight at screening, baseline and weeks 16 and 36) at clinic visits; adverse event and concomitant medication monitoring at all time points (adverse events from baseline onwards; serious adverse events from screening onwards).

ABC-16, Activities-Specific Balance Confidence Scale; ADL, activities of daily living; CSF, cerebrospinal fluid; C-SSRS, Columbia Suicide Severity Rating Scale; EQ-5D, EuroQol 5 Dimensions Questionnaire; ET, early termination; MDS-UPDRS, Movement Disorder Society-Sponsored Revision of the Unified Parkinson's Disease Rating Scale; MoCA, Montreal Cognitive Assessment; MRI, magnetic resonance imaging; PBMC, peripheral blood mononuclear cell; PDQ-39, 39-point Parkinson's Disease Questionnaire; QMA, quantitative movement assessment; SCOPA-Aut, Scale for Outcomes in Parkinson's Disease for Autonomic Symptoms.

Supplementary Figure 3. Serum BIIB094 concentration–time profile in participants receiving single IT doses in Part A

Data (chart lines) are mean and standard deviation (error bars). Dashed line represents LLOQ (1.00 ng/mL); values below this are set to 0 in Part A. Participants receiving placebo and outliers (defined as baseline value not 0 or below LLOQ, post-baseline value greater than 10,000 ng/mL, or out of 4 SD range by study drug by visit) were excluded from this analysis.

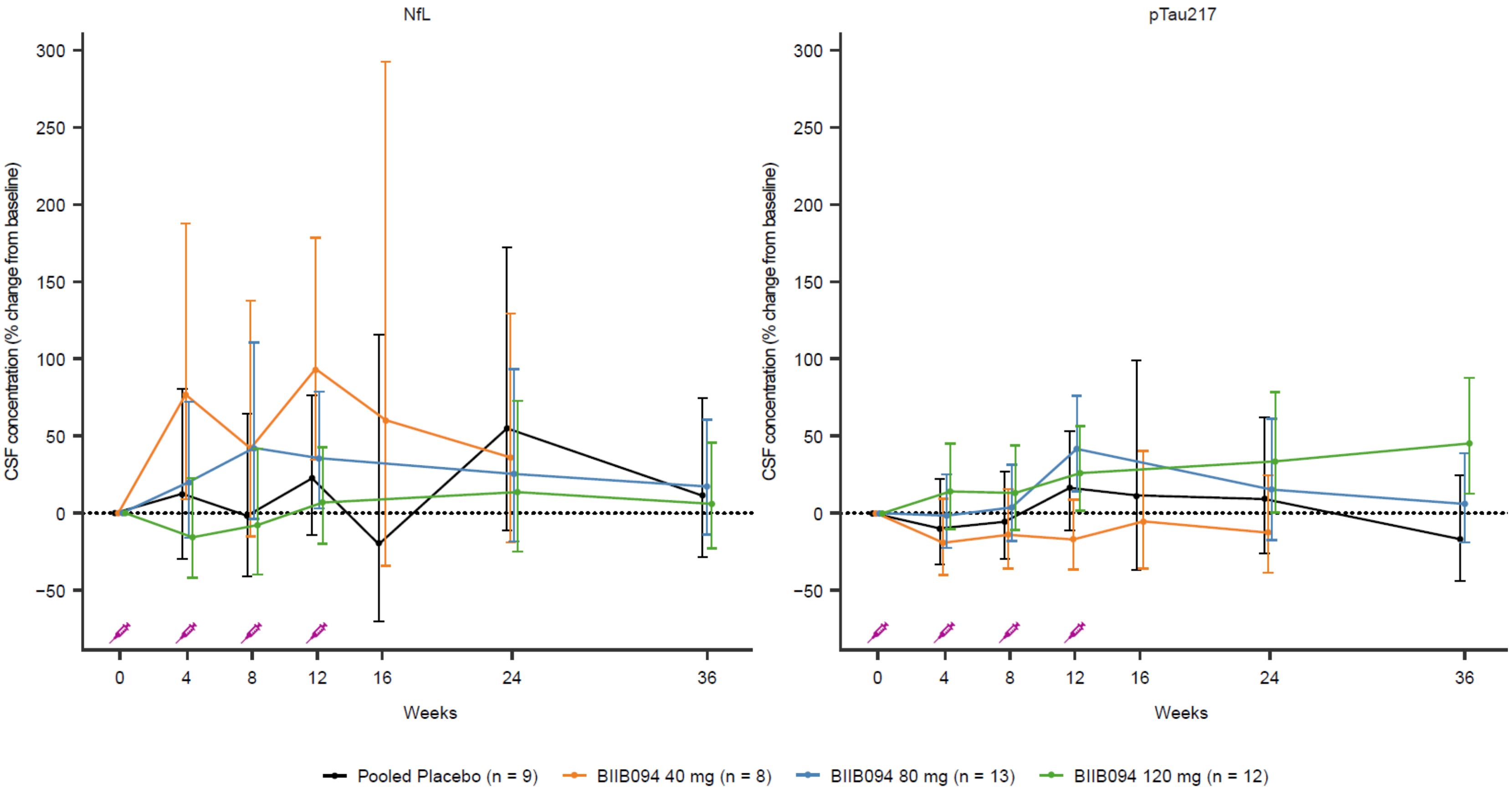
IT, intrathecal; LLOQ, limit of quantification; SD, standard deviation.



Supplementary Figure 4. CSF neurodegenerative biomarkers (NfL and pTau217) from study Part B

Arrowheads indicate the days on which the intervention (BIIBI094 or placebo) was administered. Data are pooled for the BIIB094 80 mg and 120 mg doses and for placebo. Data (chart lines) are percentage change from baseline (least squares-mean estimates) and the 95% confidence interval (error bars). Syringe symbols indicate injection of BIIB094.

CI, confidence interval; CSF, cerebrospinal fluid; NfL, neurofilament light, pTau, phosphorylated tau.

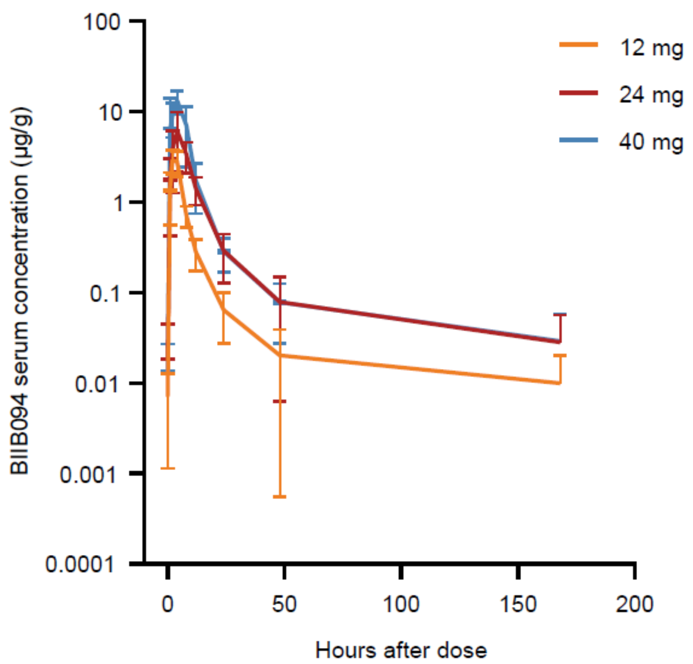


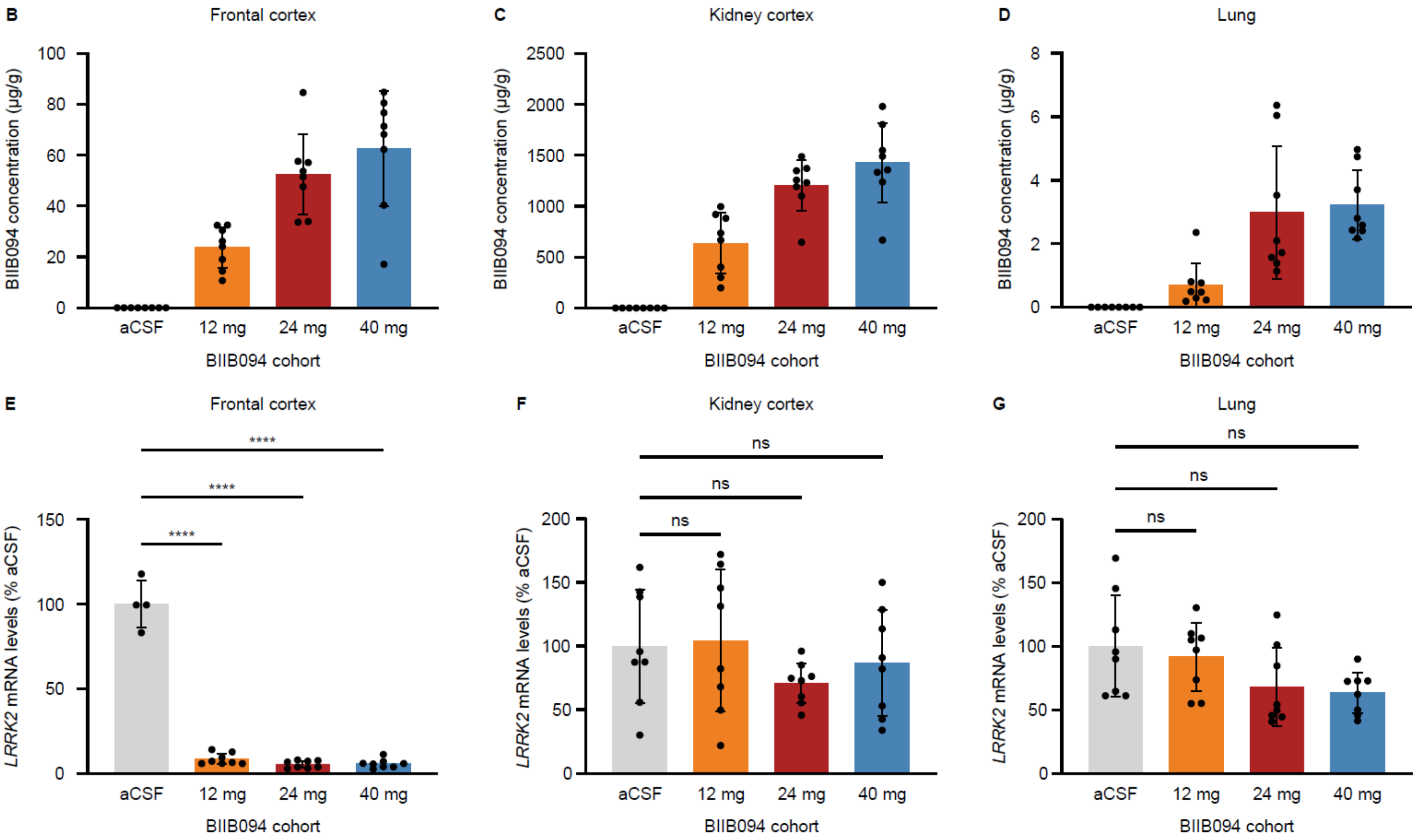
Supplementary Figure 5. BIIB094 lowers *LRRK2* mRNA in the brain but not systemically following IT dosing in cynomolgus macaques

Cynomolgus macaques (*N* = 32; *n* = 8 per cohort [4 males, 4 females]) were intrathecally dosed with 0, 12, 24 or 40 mg of BIIB094 on day 1 and weeks 4, 8, 12, 16, 20, 24, 28, 32, 36 and 40 were sacrificed at 7 days post last dose. (A) Serum BIIB094 concentration on day 281. (B–D) BIIB094 levels were evaluated in (B) frontal cortex (C) kidney cortex and (D) lung. (E–G) *LRRK2* mRNA levels were evaluated in (E) frontal cortex (F) kidney cortex and (G) lung. One-way ANOVA with Dunnett's multiple comparison. Data (charts lines or bars) are mean and standard deviation (error bars). ****: $p < 0.0001$.

aCSF, artificial cerebrospinal fluid; IT, intrathecal; ns, non-significant; *LRRK2*, *leucine-rich repeat kinase 2* gene.

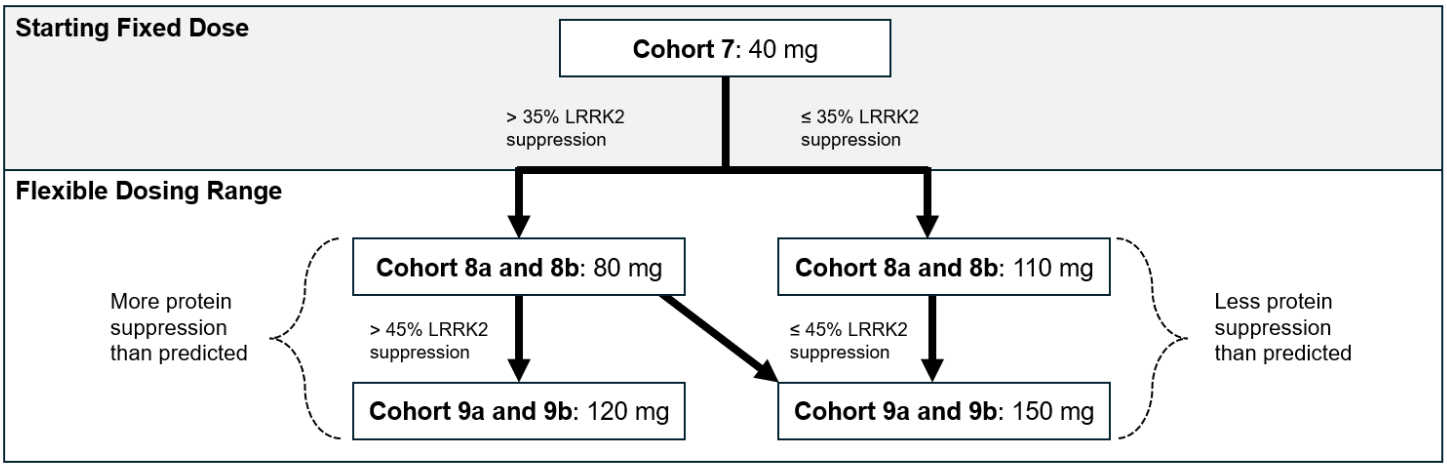
A




B
Frontal cortex
BIIB094 concentration (µg/g)
0
20
40
60
80
100
aCSF
12 mg
24 mg
40 mg
BIIB094 cohort
C
Kidney cortex
BIIB094 concentration (µg/g)
0
500
1000
1500
2000
2500
aCSF
12 mg
24 mg
40 mg
BIIB094 cohort
D
Lung
BIIB094 concentration (µg/g)
0
2
4
6
8
aCSF
12 mg
24 mg
40 mg
BIIB094 cohort
E
Frontal cortex
****
****
****
LRRK2 mRNA levels (% aCSF)
0
50
100
150
aCSF
12 mg
24 mg
40 mg
BIIB094 cohort
F
Kidney cortex
ns
ns
ns
LRRK2 mRNA levels (% aCSF)
0
50
100
150
200
aCSF
12 mg
24 mg
40 mg
BIIB094 cohort
G
Lung
ns
ns
ns
LRRK2 mRNA levels (% aCSF)
0
50
100
150
200
aCSF
12 mg
24 mg
40 mg
BIIB094 cohort

Supplementary Figure 6. Schematic of the BIIB094 dose escalation decision tree for part B (multiple ascending doses)

LRRK2, leucine-rich repeat kinase protein.



Supplementary tables

Supplementary Table 1. BIIB094 serum pharmacokinetic parameters

Part A: single ascending doses	10 mg	30 mg	40 mg	80 mg	120 mg	150 mg
AUC_{∞}: geomean (CV%), h*ng/mL [*n*]	866.87 (11) [2]	2001.73 (23) [2]	4503.68 (18) [2]	7738.11 (43) [2]	16243.31 (9) [4]	15198.36 (31) [5]
$AUC_{0\text{-last}}$: geomean (CV%), h*ng/mL [*n*]	794.64 (3) [3]	1320.29 (52) [3]	2235.58 (53) [7]	4054.64 (109) [6]	12789.99 (39) [6]	14198.01 (32) [6]
$t_{½}$: geomean (CV%), h [*n*]	3.57 (190) [2]	7.89 (38) [2]	13.50 (366) [3]	9.34 (103) [2]	21.09 (11) [4]	20.03 (31) [6]
C_{max}: geomean (CV%), ng/mL [*n*]	107.66 (102) [3]	99.52 (112) [3]	183.28 (98) [7]	351.82 (188) [6]	950.65 (121) [6]	688.07 (51) [6]
T_{max}: median (min, max), h [*n*]	1.05 (1.0, 8.2) [3]	4.20 (4.1, 24.0) [3]	3.97 (1.0, 23.6) [7]	5.98 (2.0, 24.4) [6]	2.13 (0.9, 8.3) [6]	4.07 (2.0, 8.0) [6]

Part B: multiple ascending doses	40 mg	80 mg (non-*LRRK2*-PD)	80 mg (*LRRK2*-PD)	120 mg (non-*LRRK2*-PD)	120 mg (*LRRK2*-PD)
Day 1					
$AUC_{0\text{-last}}$: geomean (CV%), h*ng/mL [*n*]	2533.28 (91) [8]	6028.32 (130) [6]	9999.02 (36) [7]	12112.46 (29) [6]	17127.69 (37) [6]
$AUC_{0\text{-tau}}$: geomean (CV%), h*ng/mL [*n*]	4526.91 (35) [4]	9235.38 (19) [2]	10616.76 (42) [5]	14670.77 (10) [4]	14673.83 (34) [3]
$t_{½}$: geomean (CV%), h [*n*]	4.01 (33) [4]	7.57 (117) [2]	14.11 (79) [5]	12.61 (67) [4]	13.27 (53) [3]
T_{max}: median (min, max), h [*n*]	5.97 (1.0, 6.2) [8]	5.05 (1.0, 23.3) [6]	4.00 (3.8, 8.0) [7]	4.13 (2.1, 8.1) [6]	2.19 (1.0, 4.2) [6]
C_{max}: geomean (CV%), ng/mL [*n*]	281.24 (232) [8]	361.86 (97) [6]	790.82 (33) [7]	1060.02 (57) [6]	2537.40 (76) [6]
Day 29					
T_{max}: median (min, max), hour [*n*]	4.00 (0.9, 5.8) [7]	3.08 (1.0, 5.9) [6]	2.07 (1.8, 6.0) [7]	5.01 (1.0, 5.9) [4]	5.02 (4.0, 6.0) [6]
C_{max}: geomean (CV%), ng/mL [*n*]	370.47 (286) [7]	551.21 (107) [6]	917.69 (89) [7]	1098.34 (76) [4]	1006.98 (92) [6]
Day 57					
T_{max}: median (min, max), h [*n*]	4.02 (2.0, 6.2) [7]	2.15 (1.0, 4.2) [5]	5.00 (2.3, 6.1) [6]	4.88 (2.3, 6.0) [4]	5.83 (4.0, 6.1) [5]
C_{max}: geomean (CV%) ng/mL [*n*]	312.31 (290) [7]	259.83 (309) [5]	585.26 (118) [6]	562.18 (115) [4]	752.70 (147) [5]
Day 85					
T_{max}: median (min, max), h [*n*]	4.12 (1.0, 6.1) [7]	4.20 (1.0, 6.0) [5]	5.82 (2.0, 6.1) [6]	2.07 (0.9, 5.9) [5]	3.92 (2.0, 5.9) [3]
C_{max}: geomean (CV%), ng/mL [*n*]	231.31 (123) [7]	389.67 (241) [5]	508.89 (82) [6]	2038.96 (75) [5]	2611.77 (35) [3]

AUC, area under the curve; AUC_{∞}, AUC from time zero to infinity; $AUC_{0\text{–last}}$, AUC from time 0 to time of the last measurable concentration; AUC_{tau}, AUC from time 0 to end of dosing interval; C_{max}, maximum observed concentration; *LRRK2*, *leucine-rich repeat kinase 2* gene; *LRRK2*-PD, *LRRK2* variant associated PD; non *LRRK2*-PD, non *LRRK2* variant associated PD; PD, Parkinson's disease; T_{max}, time to C_{max}; $t_{½}$, terminal elimination half-life.

Supplementary Table 2. Participant eligibility criteria

Inclusion criteria[a]	
Informed consent	• The ability to understand the purpose and risks of the study and provide signed and dated informed consent and authorization to use confidential health information in accordance with national and local participant privacy regulations
Age	≥ 35–80 years old at the time of informed consent
Symptomatic Parkinson's disease	• Diagnosed with PD within 7 years of initial enrollment (i.e., at time of SAD enrollment for rollover participants), without major motor fluctuations or dyskinesia that may interfere with study drug and assessments in the opinion of the investigator after consultation with the Sponsor • Participants must have at least: – One of the following: ○ Resting tremor and bradykinesia ○ Bradykinesia and rigidity ○ Rigidity and resting tremor – OR ○ Either asymmetric resting tremor or asymmetric bradykinesia
Part A (SAD) enrollment	Any eligible participant with PD regardless of *LRRK2* variant status may be enrolled in cohorts 1, 2, 3, 4, 5 and 6
Part B (MAD) enrollment	• Any eligible participant with PD, regardless of *LRRK2* variant status, may be enrolled in cohort 7 • Documentation of *LRRK2* variant status before Part B screening visit 2 is required for participants enrolling in cohorts 8a/8b and 9a/9b[b] • The pathogenic or likely pathogenic variants in the *LRRK2* gene may be considered for study inclusion are N1437H, R1441G, R1441C, R1441H, Y1699C, S1761R, G2019S and I2020T – Eligible participants with genetic test results verifying the absence of pathogenic *LRRK2* variants will be assigned to cohorts 8a and 9a • Eligible participants with genetic test results verifying the presence of a pathogenic *LRRK2* variants will be assigned to cohorts 8b and 9b
Symptom status	Modified Hoehn and Yahr Stage ≤ 3
Previous treatments	• Participants should fall into one of the following categories: – Treatment-naive and, in the opinion of the Investigator, not expected to require symptomatic therapy during the course of the study OR – On a stable dose of symptomatic therapy, which could be up to three drugs, one from each group listed as follows: ○ Selegiline, rasagiline or safinamide (up to 10 mg selegiline per 24 hours, or 1 mg rasagiline per 24 hours or 100 mg safinamide per 24 hours) ○ Carbidopa/levodopa immediate- or extended-release (up to the equivalent of 150/600 mg immediate or extended-release carbidopa/levodopa per 24 hours or the equivalent of 292.5/1170 mg carbidopa/levodopa in extended-release capsules)

	o Pramipexol, ropinirole or amantadine (up to 1.5 mg pramipexole, 6 mg ropinirole or 300 mg amantadine per 24 hours) • Participants should be on a stable dose of symptomatic therapy for a minimum of 8 weeks before screening and, in the opinion of the Investigator, not expected to require dose adjustments, additional symptomatic therapy or discontinuation of treatment throughout the duration of the study • Alternative dosing regimens using newer medications or formulations may be permitted, based upon discussion between the Investigator and the Study Medical Director to clarify eligibility
Contraception	• For women of childbearing potential and all men, willingness to practice highly effective contraception during and for 5 months after their last dose of study drug • Participants should not donate sperm or eggs during the study and for ≥ 5 months after their last dose of study drug
Exclusion criteria[a]	
Medical history	• MoCA score < 23, dementia or other significant cognitive impairment that, in the opinion of the Investigator, would interfere with study evaluation • History of any brain surgery for PD (e.g., pallidotomy, deep brain stimulation, or fetal tissue transplant) or history of focused ultrasound treatment at any time; or history of neuromodulation procedures, including, but not limited to, TMS, tDCS/tACS that have been performed within 90 days of screening • Known or suspected cause of parkinsonism other than neurodegenerative PD: Participants with atypical parkinsonian syndromes (e.g., Lewy body dementia, progressive supranuclear palsy or multiple system atrophy, presence of drug-induced parkinsonism (e.g., neuroleptics, metoclopramide, flunarizine and so on), metabolic identified neurogenetic disorders (e.g., Wilson's disease) or encephalitis • Any contraindications to having a brain MRI (e.g., pacemaker; MRI-incompatible aneurysm clips, artificial heart valves or other metal foreign body; claustrophobia that cannot be medically managed without requiring general anesthesia, etc.) • Any contraindications to LP procedures, including but not limited to: – Known or suspected structural abnormality of the lumbar spine, including but not limited to X-ray, MRI or myelographic evidence of significant lumbar spine abnormalities or other anatomical factors at or near the LP site that, in the opinion of the Investigator, may interfere with the performance of the LP, rend aneurysm clips, artificial heart valves or other metal foreign body; claustrophobia that cannot be medically managed without requiring general anesthesia, etc.) – Presence of risk for increased or uncontrolled bleeding and/or risk of bleeding that is not managed optimally and might place a participant at an increased risk for intraoperative or postoperative bleeding. These could include, but are not limited to, anatomical factors at or near the LP site (e.g., vascular abnormalities, neoplasms or other abnormalities), known underlying disorders of the coagulation cascade, platelet function or platelet count (e.g., hemophilia, von Willebrand's disease, liver disease) • Unstable psychiatric illness, including psychosis, suicidal ideation or untreated major depression within 90 days before screening, as determined by the Investigator • History or screening MRI results showing evidence of structural abnormalities that could contribute to the participant's clinical state or any finding that might pose a risk to the participant • Transient ischemic attack or stroke or any unexplained loss of consciousness within 1 year before screening • History of unstable angina, myocardial infarction, chronic heart failure (New York Heart Association Class 3 or 4) or clinically significant conduction abnormalities (e.g., unstable atrial fibrillation) within 1 year before screening • Chronic, uncontrolled hypertension (average of three SBP or DBP readings at screening > 164 or ≥ 100 mmHg, respectively; any documented SBP/DBP reading > 180 or ≥ 100 mmHg, respectively, within the 3 months before day 1) • Symptomatic orthostatic hypotension despite optimal medical management such that the participant's standing mean arterial pressure at screening is

	below 75 mmHg or, in the opinion of the Investigator, would interfere with study participation • Clinically significant (as determined by the Investigator) 12-lead ECG abnormalities, including confirmed demonstration of corrected QT interval using the Fridericia formula of > 460 msec (men) and > 470 msec (women) before study drug administration • History of malignancy or carcinoma. At sponsor’s discretion, exceptions may be made for participants with a history of basal cell or squamous cell carcinomas of the skin that have been completely excised and are considered cured • Poorly controlled diabetes mellitus, as defined by having dosage adjustment of diabetic medication within 3 months before dosing (day 1) or glycosylated hemoglobin value ≥ 8% at screening • Hypersensitivity to anesthetics that will be used for the LP per institutional practice • History of any clinically significant renal or pulmonary disease • History of any clinically significant urologic, endocrinologic, hematologic, hepatic, immunologic, metabolic, neurologic, dermatologic, ischemic or other major diseases, as determined by the Investigator
Infection	• History or positive test result at screening for HIV • History or positive test result at screening for hepatitis C virus antibody • Current hepatitis B virus (HBV) infection (defined as positive for HbsAg and/or hepatitis B core antibody [anti-HBc]). Participants with immunity to hepatitis B from previous natural infection (defined as negative HbsAg, positive anti-HBc, and positive hepatitis B surface antibody [anti-HBs]) or vaccination (defined as negative HbsAg, negative anti-HBc and positive anti-HBs) are eligible to participate in the study • Currently active infection or serious infection (e.g., pneumonia, septicemia) within 8 weeks before day 1, as determined by the Investigator
Laboratory values	• Liver function test results, direct bilirubin, total bilirubin or creatinine levels ≥ 1.5 times the ULN at the screening visit. Participants with abnormal test results may be rescreened one time at the discretion of the Investigator. Participants with clinically nonsignificant out-of-range laboratory results may be enrolled at the discretion of the Investigator after consultation with the Medical Monitor and/or Study Medical Director – Participants with previously diagnosed Gilbert’s syndrome and elevated levels of bilirubin consistent with such diagnosis may be allowed in the study • Estimated glomerular filtration rate (calculated according to the Modification of Diet in Renal Disease equation) < 60 mL/min at screening • Screening value for hemoglobin < 12 g/dL for men or < 11 g/dL for women • Platelet count value that is below the lower limit of normal, or screening values of INR, PT, or aPTT that are not within normal ranges
Previous and concomitant medication	• Any live, attenuated immunization/vaccination within 2 months before the first dose of study drug and for the duration of the study or plans for any immunization/vaccination during the 10 days before and after dosing visits • Use of any of the following drugs within 180 days before day 1 and for the duration of the study: typical or atypical antipsychotics (including, but not limited to, quetiapine, pimavanserin, clozapine, olanzapine, flunarizine and aripiprazole), metoclopramide or alpha methyldopa • Use of any of the following drugs within 90 days before day 1 and for the duration of the study: methylphenidate, cinnarizine, tetrabenazine, reserpine, amphetamine, memantine, cholinesterase inhibitors (rivastigmine, donepezil, galantamine and tacrine), monoamine oxidase type A inhibitors (pargyline, phenelzine and tranylcypromine) • Use of any glucagon-like peptide-1 (GLP-1) agonists (e.g., exenatide, liraglutide, lixisenatide, albiglutide and dulaglutide) within 90 days before day 1

	and for the duration of the study • Use of immunosuppressive drugs (including systemic corticosteroids) within 90 days before day 1 and for the duration of the study. Short-term corticosteroids for the treatment of reversible conditions and local corticosteroids may be permitted, following a consultation with the Sponsor • Treatment with parenteral immunoglobulin therapy, blood products, plasma derivatives, plasma exchange or plasmapheresis within 90 days before day 1 and for the duration of the study • Participation or planned enrollment in any other clinical study or treatment with any investigational drug or approved therapy for investigational use within 30 days (or 5 half-lives, whichever is longer) before screening and for the duration of the study • Use of nilotinib or Mucuna pruriens within 90 days before day 1 and for the duration of the study • Treatment with antiplatelet or anticoagulant therapy ≤ 14 days before screening (with the exception of aspirin ≤ 100 mg/day) or anticipated use during the study, including but not limited to clopidogrel, dipyridamole, warfarin, heparin or heparinoids, dabigatran, rivaroxaban and apixaban • Use of allowed medications not previously specified at doses that have not been stable for at least 8 weeks before day 1, and/or that are not expected to remain stable for the duration of the study – Routine insulin medication adjustments for diabetic participants are permitted
General or other	• Participation or planned enrollment at another site activated for this study • Employment at Biogen or at sites activated for this study. Immediate relatives of Biogen or site employees are also excluded from study participation • Participation in concurrent noninterventional studies unless specifically authorized by the Sponsor • History of drug or alcohol abuse within the past 5 years (as defined by the Investigator), a positive urine drug test at screening, or an unwillingness to abstain from these substances for the duration of the study – Participants who test positive for cannabinoids due to occasional marijuana use or CBD oil, as determined by the Investigator, and who agree to refrain from use for the duration of the study may be enrolled at Investigator's discretion, after a consultation with the Sponsor • Current (within 6 months) smoking of cigarettes or usage of any other tobacco-containing products • Surgery within 12 weeks before day 1 (other than minor cosmetic surgery and minor dental surgery, as determined by the Investigator) • Female participants who are pregnant, currently breastfeeding or attempting to conceive during the study • Blood donation (one unit or more) within 8 weeks before day 1 (must also refrain from donating blood for the duration of the study) • Unwillingness or inability to comply with study requirements, including the presence of any condition (physical, mental or social) that prevents the participant from taking part in visits as scheduled • Other unspecified reasons that, in the opinion of the Investigator or Sponsor, make the participant unsuitable for enrollment

[a]At screening, or at the time point specified in the individual eligibility criterion listed.
[b]For *LRRK2* variant status documentation to be considered acceptable, it must be obtained from an accredited laboratory using a certified test for the status (i.e., positive/negative) for each of the eight pathogenic variants listed. If only a subset of the eight variants was assayed using a certified test, positive *LRRK2* variant status will be sufficient for inclusion in cohorts 8b/9b; however, an *LRRK2*-negative result on a partial test would not suffice. These accreditations can be Clinical Laboratory Improvement Amendments, College of American Pathologists or at a local laboratory that has undergone a similar rigorous review. The *LRRK2* variant status of all participants will be verified using a centralized laboratory in the US as part of the screening process.
aPTT, activated partial thromboplastin time; CBD, cannabidiol; DBP, diastolic blood pressure; ECG, electrocardiogram; GLP-1, glucagon-like peptide-1; HbsAq, hepatitis B surface antigen; HIV, human immunodeficiency virus; INR, international normalized ratio; LRRK2, leucine-rich repeat kinase 2; LP, lumbar puncture; MAD, multiple ascending dose; MoCA, Montreal Cognitive

Assessment; MRI, magnetic resonance imaging; PD, Parkinson’s disease; PT, prothrombin time; SAD, single ascending dose; SBP, systolic blood pressure; tDCS/tACS, transcranial direct or alternating current stimulation; TMS, transcranial magnetic stimulation; ULN, upper limit of normal.

Supplementary Table 3. Projected LRRK2 suppression at each planned dose in Part B

Cohort	Dose (mg)	Biomarker Assessment Timing	CSF LRRK2 Mean Suppression Criterion	Prespecified Next-Dose Action	Notes
7	40	After completion of Cohort 7	> 35%	Escalate to 80 mg (Cohorts 8a/8b)	Safety/PK/tolerability review applies
7	40	After completion of Cohort 7	≤ 35%	Escalate to 110 mg (Cohorts 8a/8b)	Safety/PK/tolerability review applies
8a/8b	If on 80	At time of 4th 80 mg dose	> 45%	Escalate to 120 mg (Cohort 9a or 9b)	Safety/PK/tolerability review applies
8a/8b	If on 80	At time of 4th 80 mg dose	≤ 45%	Escalate to 150 mg (Cohort 9a or 9b)	Safety/PK/tolerability review applies

CSF, cerebrospinal fluid; LRRK2, leucine-rich repeat kinase 2 protein; PK, pharmacokinetics.

Supplementary Table 4. Study objectives

Primary objective	Primary endpoints
To evaluate the safety and tolerability of single and multiple doses of BIIB094	• Incidence of AEs, assessed from day 1 to final study visit • Incidence of SAEs assessed from screening to final study visit
Secondary objectives	**Secondary endpoints**
To evaluate the PK profile of BIIB094	• Serum concentrations of BIIB094 • Serum PK parameters: – AUC from time zero to infinity (AUC_{∞}) – AUC from time 0 to time of the last measurable concentration ($AUC_{0\text{-last}}$) – Maximum observed concentration (C_{max}) – Time to C_{max} (T_{max}) – Terminal elimination half-life ($t_{½}$)
Exploratory objectives	**Exploratory endpoints**
To evaluate the PK of BIIB094 in CSF	• CSF trough concentrations of BIIB094
To evaluate the effects of BIIB094 on levels of potential biomarkers	• Changes from baseline in CSF levels of LRRK2 protein and/or measures of LRRK2 substrates • Changes from baseline in CSF markers of neurodegeneration (may include but are not limited to α-syn, tau, neurofilament, and inflammatory markers) • Changes from baseline in LRRK2 protein, LRRK2 substrates and/or neurodegeneration biomarker levels in other matrices, which may include serum, plasma, urine, PBMCs and RNA • For Part B (MAD): Changes from baseline in neuroimaging measures, which may include quantitative magnetic resonance imaging MRI) metrics (including but not limited to neuromelanin-sensitive and diffusion MRI)
To evaluate measures of clinical efficacy and quality of life in relationship to BIIB094 and biomarkers	• Motor and cognitive symptoms of PD as measured by the MDS-UPDRS, the Modified Hoehn and Yahr scale, QMA and MoCA • Motor function and related ADL, as measured by the Modified Schwab and England ADL scale • Nonmotor somatic aspects of PD, as measured by the SCOPA-Aut • Health-related quality of life and mental health, as measured by PDQ-39, EQ-5D and ABC-16

ABC-16, Activities-Specific Balance Confidence Scale; ADL, activities of daily living; AE, adverse event; AUC, area under the curve; CSF, cerebrospinal fluid; EQ-5D, EuroQol 5 dimensions questionnaire; *LRRK2, leucine-rich repeat kinase 2* gene; MAD, multiple ascending dose; MDS-UPDRS, Movement Disorder Society Sponsored Revision of the Unified Parkinson's Disease Rating Scale; MoCA, Montreal Cognitive Assessment; MRI, magnetic resonance imaging; PBMC, peripheral blood mononuclear cell; PD, Parkinson's disease; PDQ-39, 39-point Parkinson's Disease Questionnaire; PK, pharmacokinetic; QMA, quantitative movement assessment; RNA, ribonucleic acid; SCOPA-Aut, Scale for Outcomes in Parkinson's Disease for Autonomic Symptoms; SAE, serious adverse event.

Supplementary Table 5: PRM target list

Protein	Peptide	Mass, m/z	(N)CE
Ubiquitin	TITLEVEPSDTIENVK	894.4744	22
Ubiquitin	TITLEVEPSDTIENVK$_{13C(6)15N(2)}$	898.4744	22
Cathepsin D	VSTLPAITLK	521.8290	22
Cathepsin D	VSTLPAITLK$_{13C(6)15N(2)}$	525.8361	22
Glucocerebrosidase	LLMLDDQR	502.2657	20
Glucocerebrosidase	LLMLDDQR$_{13C(6)15N(2)}$	507.2698	20
GPNMB	AYVPIAQVK	494.7949	20
GPNMB	AYVPIAQVK$_{13C(6)15N(2)}$	498.8020	20
LAMP2	IPLNDLFR	494.2847	28
LAMP2	IPLNDLFR$_{13C(6)15N(2)}$	499.2889	28
LAMP1	AFSVNIFK	463.2607	21
LAMP1	AFSVNIFK$_{13C(6)15N(2)}$	467.2678	21
Cathepsin B	DIMAEIYK	491.7493	20
Cathepsin B	DIMAEIYK$_{13C(6)15N(2)}$	495.7564	20
GM2A	EVAGLWIK	458.2686	24
GM2A	EVAGLWIK$_{13C(6)15N(2)}$	462.2757	24
Cathepsin F	FSDLTEEEFR	636.7908	22
Cathepsin F	FSDLTEEEFR$_{13C(6)15N(2)}$	641.7949	22
Cathepsin L	NSWGEEWGMGGYVK	800.3485	24
Cathepsin L	NSWGEEWGMGGYVK$_{13C(6)15N(2)}$	804.3556	24
LC3	IPVIIER	420.2711	28
LC3	IPVIIER$_{13C(6)15N(2)}$	425.2752	28
Cathepsin S	GIDSDASYPYK	608.2800	28
Cathepsin S	GIDSDASYPYK$_{13C(6)15N(2)}$	612.2871	28

GM2A, ganglioside GM2 activator; GPNMB, glycoprotein non-metastatic protein B; LAMP, lysosomal-associated membrane protein; LC3, microtubule-associated protein 1A/1B-light chain 3; m/z, mass to charge; N(CE), normalized collision energy; PRM, parallel reaction monitoring.